

Bridging RiC-O and PREMIS: An Alignment Framework for Digital Archives

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1 Introduction

Digital records have become the primary medium through which modern organizations, government institutions, and individuals document their activities, and preserve cultural memory [4, 20, 18, 15, 24]. Unlike traditional paper-based archives, digital archives are not static physical repositories that hold fixed, human-readable documents. Instead, digital records exist as dynamic, software-dependent, and complex digital objects. The primary motivation for digital archives today extends far beyond simple long-term storage [12, 13]; it depends on technical and organizational conditions such as file formats, software, hardware, storage environments, integrity checks, preservation actions, rights, and the evidence documenting those actions. In this context, static, traditional metadata models do not suffice. Relational schemas and hierarchical, document-centric XML structures struggle to capture the highly interconnected nature of digital preservation environments. To ensure long-term authenticity, integrity, and intellectual accessibility across evolving technology stacks, digital archives require Linked Data models based on Semantic Web standards [1, 2].

To capture the multidimensional nature of archival information, the archival community, through the International Council of Archives (ICA), developed Records in Contexts (RiC). RiC was developed to move archival description away from isolated hierarchical descriptions toward an entity-centric and relationship-oriented representation of records, agents, activities, events, functions, and other contextual entities [21]. RiC-CM [21, 16] provides the conceptual model, while RiC-O [17] expresses its concepts through RDF and OWL, making them suitable for Linked Data environments. This semantic orientation is particularly relevant to digital archives because information can be distributed across heterogeneous systems and benefit from explicit links between records, actions, people, software, policies, and evidence. However, while RiC-O excels at modeling historical, organizational, and contextual provenance, it lacks explicit, fine-grained semantic

constructs to model operational digital preservation microservices and technical object characteristics.

On the other hand, the PREMIS Data Dictionary for Preservation Metadata [23] is the widely-used international standard for documenting the preservation metadata of digital objects, including the operational lifecycle of digital objects, preservation events, rights, and technical dependencies. While PREMIS provides detailed technical granularity, it relies on external systems to supply the archival descriptions that give preserved objects their evidential meaning.

When focusing on archives of digital records, the two models, RiC and PREMIS, consequently address complementary dimensions of the same information environment, where RiC emphasizes archival context, while PREMIS emphasizes preservation management. This constitutes the main problem addressed by this paper. In particular, this paper focuses on examining how to leverage the semantic expressiveness of Linked Data (specifically, RiC-O) alongside the operational preservation granularity of PREMIS to create a unified, machine-readable knowledge graph. If archival description and preservation metadata remain isolated, a repository may know that a record exists and who created it, while storing preservation events and technical evidence elsewhere without a direct semantic bridge. Conversely, a preservation system may record a checksum, migration, or software dependency without representing the archival context that gives the preserved object evidential meaning.

To address this limitation, in this paper, we propose a lightweight bridge ontology that formalizes semantic relationships and class alignments between RiC-O and PREMIS v3, focusing on digital records and archives. Our approach provides a formal, machine-processable bridge without replacing PREMIS or refactoring RiC-O, also specifying instance-level multi-typing relationships between certain core structural entities. This approach preserves the native semantics of both domain standards while enabling the implementation of a unified Knowledge Graph allowing querying across archival descriptions and technical preservation metadata, as validated through concrete digital repository scenarios. The rest of this paper is organized as follows: Section 2 reviews the related work. Section 3 provides the necessary formal background on OWL ontologies, RiC-CM/RiC-O, and PREMIS v3. Section ?? details the proposed extension and bridging architecture, and Section ?? concludes the paper with directions for future work.

2 Related Work

The literature on semantic interoperability in archives increasingly treats archival models as knowledge representation frameworks rather than as fixed descriptive forms. RiC-CM responds to the limitations of traditional standards by representing records through entities and relationships and by supporting contextual networks [14, 8]. The approach has also been examined in relation to CIDOC CRM, demonstrating that conceptual correspondences can support interoperability across cultural heritage systems [3, 22]. Such work is important for the

present study because it shows that RiC-CM can participate in mappings across models without requiring a single vocabulary to replace all others.

RiC-O further strengthens this potential by implementing RiC-CM through Semantic Web technologies. Clavaud and Wildi’s work demonstrated the role of RiC-O in Linked Open Data [6], while the RiC-O Converter [5] showed that existing EAD and EAC-CPF descriptions can be transformed into RDF datasets compatible with RiC-O. Applications of RiC to concrete archival collections (e.g., [10]) have also shown its capacity to model complex relationships between records, agents, activities, and historical contexts. These developments suggest that RiC is suitable as a semantic backbone for archival information, including specialized extensions.

PREMIS [23, 11, 9, 25, 7] addresses a different layer of the information environment. Its Data Dictionary defines preservation metadata required to maintain digital objects over time, including technical characteristics, preservation events, agents, rights, and integrity information [23]. Practical implementation studies have shown the usefulness of PREMIS for tracking preservation processes, while PREMIS OWL has extended the standard toward semantic representation [7]. The development of semantic representations is especially relevant because both RiC-O and PREMIS can be discussed within an RDF/OWL environment rather than only as independent XML or database structures.

The literature nevertheless leaves an integration gap. Existing research has examined archival interoperability, PREMIS implementation, and semantic representations separately, but the thesis underlying this paper identifies no established systematic mapping that combines RiC-O’s contextual expressiveness with PREMIS’s preservation specificity. This gap motivates a mapping based on semantic function and a targeted extension rather than a wholesale redesign of RiC-CM.

3 Preliminaries

Ontologies provide formal vocabularies for representing concepts, properties, and relationships in a domain (Gruber, 1993). In the Semantic Web architecture, RDF represents information as subject-predicate-object triples, RDFS supports classes and hierarchies, and OWL provides richer logical constructs for equivalence, specialization, restrictions, and reasoning (W3C 2014a; W3C 2014b; W3C 2012). These technologies matter for archival interoperability because mappings can be expressed explicitly and machine-processably. A semantic relation can therefore distinguish equivalence from specialization or documentation instead of treating two similarly named metadata fields as automatically identical.

Ontology has its roots in philosophy, where it examines the meaning of existence as well as the very nature of beings. However, in computer science, especially in knowledge engineering, the term takes on a more technical character: ontology means a formal, explicitly formulated representation of a domain of knowledge. Gruber describes it simply as “an explicit specification of a conceptual representation” [GRU93]. In other words, an ontology maps the basic concepts

of a domain, defines the relationships between them, records their properties, and establishes the rules that govern them [AGH+25]. Ontologies play a crucial role when different information systems need to collaborate. Thanks to a shared vocabularies and the clear organization of knowledge, a unified understanding of information becomes possible. They are used in many fields, such as artificial intelligence, knowledge systems, biomedical applications, information retrieval systems, IoT ecosystems, and in the knowledge graphs used by companies like Google and IBM.

An ontology O is formally defined as the tuple $O = \langle C, R, I \rangle$, where C is the set of Classes / Concepts, R is the set of Relations / Properties, I is the set of Individuals / Instances of classes in C . A class represents abstract groups, categories, or types of objects belonging to a domain (e.g., Person, Device, Process, Activity, Object). Individuals or instances typically describe the ground-level objects or tokens that populate the classes (e.g., *sensor*₀₁ as an instance of Device). In essence, when an object or entity is of a given class, we say that it is an instance of that class. A relation/property is a binary relationship that link two classes in C , together. Properties are divided into two types: Object properties (that link an instance of one class to an instance of another class - e.g., *hasComponent*) and Data properties (that link an instance of a class to a literal data value (e.g., *hasAge* a integer). To use multiple ontologies, one approach is to use a Bridge Ontology [19] which acts as a decoupled semantic mediation layer that imports the necessary namespaces of the underlying ontologies and defines alignment rules without refactoring the underlying ontologies.

3.1 Records in Context - Conceptual Model and Ontology

The RiC-CM (Records in Contexts – Conceptual Model), developed by the International Council on Archives (ICA), seeks to integrate traditional archival description standards, such as ISAD(G), ISAAR(CPF), ISDF, and ISDIAH, within a unified conceptual framework [?].

Unlike older hierarchical models, the RiC-CM is based on an entity-centric approach. Here, archival information is organized around entities, attributes, and relationships. Its basic idea is that records are not isolated entries, but rather parts of a broader contextual framework involving people, organizations, functions, events, and processes related to their creation, management, and use. This approach enables archival information to be represented in a more multidimensional way and paves the way for data interlinking in Semantic Web and Linked Open Data environments [ICA21].

The basic entities of RiC-CM are the Record or Record Set, which corresponds to the archival record itself or a set of records (e.g., collections, fonds, series, folders). The Agent, is the person, organization, or group that creates, manages, or uses the records. The Function, which relates to activities or responsibilities and is directly linked to the creation of records and the Event, which includes actions and processes such as the creation, transfer, modification, or preservation of records. The model is complemented by entities such as

Place and Date, which help to comprehensively capture the historical and functional context of the records [ICA21]. Each entity is accompanied by specific attributes that provide detailed information about its characteristics. For example, the Agent entity includes attributes such as the agent’s identifier, name, type, dates of existence, and administrative or biographical history. Similarly, the Record entity has details such as the title, creation date, language, format, extent, level of description, and technical characteristics of the digital object. These attributes function as semantic units of information and enable the accurate documentation of archival resources [ICA21]. Figure 1 depicts the main hierarchy of the RiC-CM classes.

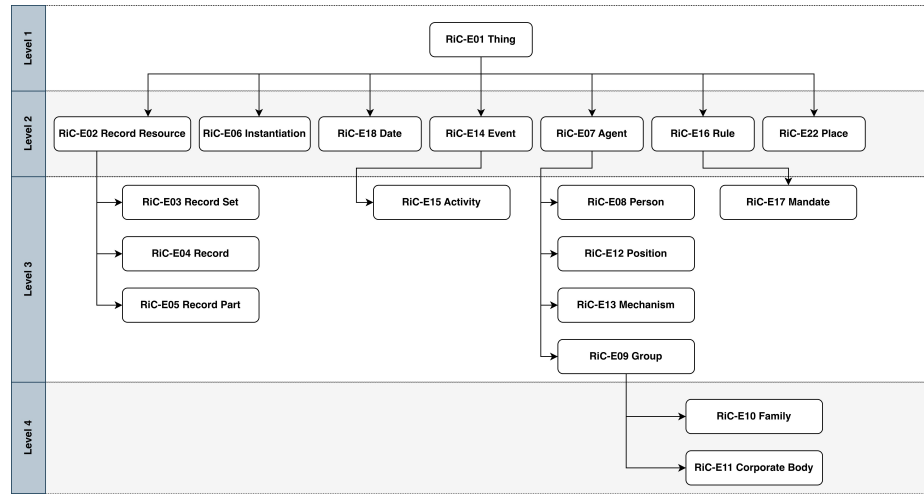


Fig. 1. RiC-CM - Class Hierarchy.

In RiC-CM, relations (i.e., relationships between entities) are of particular importance, as they serve as the primary means by which entities are linked and the archival framework is represented. Relations are used to indicate, for example, that an Agent creates or manages a Record, that a Record documents a specific Function, or that an Event affects, changes, or transforms a Record. At the same time, relations allow different sets of records, entities, and events to be linked, resulting in the creation of a dynamic information network. This network captures not only the content but also the historical, administrative, and operational context of the records [ICA21]. In Figure 2, we can see the main relations defined by the RiC-CM between the classes.

3.2 PREMIS (Preservation Metadata: Implementation Strategies)

PREMIS is the best-known metadata standard for digital preservation. It was designed to support the long-term management and preservation of digital objects

[PEC15]. The PREMIS Data Dictionary for Preservation Metadata provides a structured framework for recording technical, administrative, and management metadata related to the preservation of digital material. Many digital repositories, libraries, archives, and cultural heritage organizations implement it, resulting in a wide range of applications. Unlike general archival description models, PREMIS focuses exclusively on the needs of digital preservation, prioritizing the authenticity, integrity, accessibility, and long-term usability of digital objects [LAV14].

The conceptual structure of PREMIS is based on four core entities: Object, Event, Agent, and Rights. Figure 3 illustrates the core PREMIS entities and show how they can be connected (through relationships) with each other. The Object entity refers to the digital object itself, which can be individual files, bitstreams, or even more complex digital objects. In particular, an Object could be either an Intellectual entity (representing the conceptual entity, e.g., specific book, software), a representation of an intellectual entity (e.g., a collection of image files, a pdf or e-pub document), and a file or a bitstream (representing mainly the physical object). The Event entity is used to record every action or process that affects the digital object, such as migration, normalization, validation, replication, or integrity checking. The Agent describes the individuals, systems, or organizations that undertake or manage preservation activities. As for Rights, this is where the rights and permissions associated with preservation activities as well as access to digital objects are recorded [PEC15]. Each entity has its own specialized semantic units (i.e., attributes) that help accurately document the digital preservation process. For example, the Object entity includes attributes such as `objectIdentifier`, `objectCategory`, `compositionLevel`, `fixity`, `size`, `format`, and `creatingApplication`. Fixity is particularly valuable as it verifies integrity using checksums and hashing algorithms. Note here that PREMIS ontology is also extended through the Preservation Vocabulary¹ provided by Library of Congress; e.g., the Event Type `creation` is defined as a subclass of the class `premis:Event`.

The Event entity includes elements such as `eventIdentifier`, `eventType`, `eventDateTime`, `eventOutcome`, and `eventDetail`, allowing every preservation action and its results to be recorded in detail. The Agent has attributes such as `agentIdentifier`, `agentName`, and `agentType`. The Rights entity records permissions, restrictions, and permitted maintenance actions with properties such as `rightsBasis`, `act`, and `restriction` [PEC15]. Relationships between entities are a fundamental aspect of PREMIS, as they link objects, actions, and responsible entities within a comprehensive digital preservation framework. For example, an Event is linked to a specific Object, indicating that a migration or validation was applied to it. At the same time, an Agent is linked to the Event as the person responsible for performing that action, while the Rights entity determines whether the Agent has the right to perform actions such as copying, converting, or reproducing the digital object. Through these relationships, PREMIS creates a detailed

¹ See <http://id.loc.gov/vocabulary/preservation>

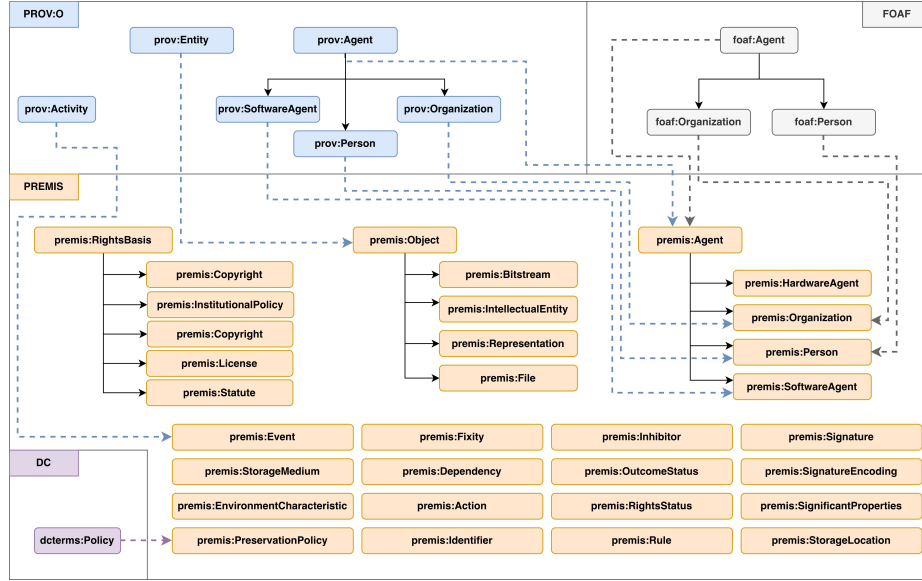


Fig. 2. Overview of PREMIS.

information network that documents every stage of the digital object’s lifecycle [PEC15]; [LAV14].

PREMIS model is given in multiple representations/serializations, such as XML Schema (for traditional structured records) and OWL / RDF ontologies (for Linked Data implementations) . Figure 2 illustrates the classes of PREMIS ontology, as well as their subclass relations. Notice that PREMIS ontology has been defined in terms of other well-established ontologies such as PROV-O, FOAF and Dublin Core. PROV-O is used to model provenance and execution history, while FOAF (Friend of a Friend) is leveraged to represent human agents and organizations, and Dublin Core is mainly used to define Preservation Policies.

4 Bridging RiC-O and PREMIS: Semantic Alignment Framework

This section focuses on presenting the interoperability framework designed to bridge the RiC-O and PREMIS (v3) ontologies. Let us initially understand the context and need for such an alignment by using a simple example. Consider a born-digital record issued on 2008-10-14, by the Ministry of Interior (`ex:Ministry_Interior`) to document the administrative decision approving a municipal infrastructure funding allocation (`ex:Approve_Funding`). The record is originally created in legacy Word format (`infra_funding.doc`) and stored for long-term archival preservation. The document was later migrated to DOCX for-

mat (`infra_funding.docx`), on 2024-09-10, to ensure compatibility with modern technology. Utilizing RiC-O captures the archival description, linking the record (`ex:Infra_2008_412`) to its creation date, its creator, the approval activity it documents etc. Concurrently, PREMIS records operational preservation metadata, including file size, SHA-256 fixity fingerprints, and the automated format migration event (`ex:Migration_DOCX`) executed by a software agent (`ex:LibreOffice_v7.5`).

To represent the aforementioned example through an RDF data graph, we depict a representative subset of this graph in Figure 3. The white-colored nodes represent instances of RiC-O classes, the orange-colored nodes represent instances of PREMIS classes, and the dual-colored nodes represent multi-typed entities, as we will describe later in this section. Through this example, we can easily see that bridging both ontologies is essential because while RiC-O captures the description of the records and treats the migration log itself as a newly documented record (`migration_logs`), PREMIS provides the detailed technical evidence of the migration, and only their alignment guarantees that the modern `.docx` file remains authentic to its 2008 `.doc` origin.

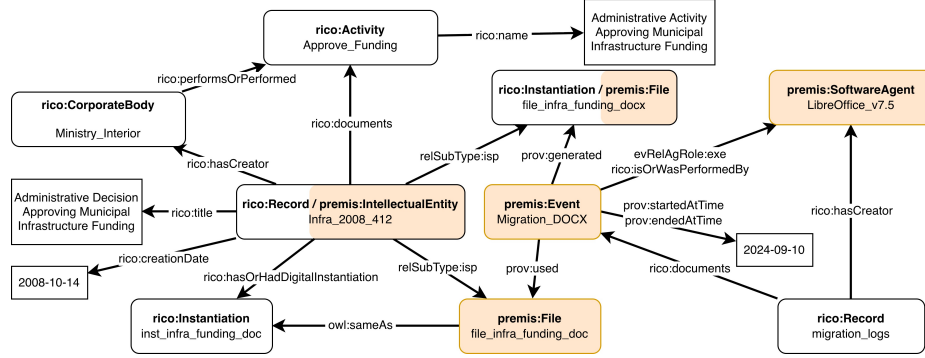


Fig. 3. Example - PREMIS & RiC-O.

To address this challenge, we focus on digital archives and records and present an alignment framework between RiC-O and PREMIS ontologies. In this work, we mainly handle semantic integration at the entity (class) level, leaving the exhaustive property alignment for future work. In particular, we propose a mapping of PREMIS classes to RiC-O ones and analyze its validity. Such a mapping is given through a lightweight, specialized Bridge Ontology. This bridge is defined by focusing on how PREMIS classes are aligned with RiC-O entities (not aiming to align all the RiC-O classes with PREMIS ones). Subsumption and equivalence relations of classes are established using `rdfs:subClassOf` and `owl:equivalentClass`, while entity reconciliation across structural instances is achieved via instance-level multi-typing using `rdf:type`. We also consider the OWL property `owl:sameAs` to specify instance-level equivalences, enabling

the reconciliation of pre-existing, independent entities generated across distinct archival and preservation repositories.

Directly modifying the source ontologies or asserting global class subsumptions across all core structural concepts risks creating ontological noise and unwanted inferential side effects. For instance, declaring a global subclass relationship between PREMIS digital objects and RiC-O record resources can force low-level, temporary bitstreams or technical utilities into the archival provenance hierarchy, diluting the evidential value of archival description. To avoid altering native specifications or forcing unnatural subsumptions, we propose a lightweight, specialized Bridge Ontology [?]. A bridge ontology acts as a decoupled semantic mediation layer that imports the necessary namespaces of both standards and formalizes precise alignment rules without refactoring underlying schemas. This strategy preserves domain autonomy, allowing preservation microservices and archival platforms to retain native vocabularies. Furthermore, by decoupling schema-level TBox axioms from instance-level ABox multi-typing, reasoners can execute queries across unified knowledge graphs while preventing invalid logical inferences. The bridge architecture also ensures long-term extensibility as future iterations of PREMIS or RiC-O evolve.

Our alignment methodology relies on a qualitative, function-driven semantic analysis rather than superficial lexical string matching. Each concept pair was evaluated based on its formal scope, structural positioning, domain and range constraints, and operational utility within digital repositories. The resulting framework categorizes relationships into three formal typologies: Direct Class Equivalence (`owl:equivalentClass`) where concepts share identical semantic boundaries; Domain Specialization (`rdfs:subClassOf`) where a PREMIS entity represents a specialized subset of a broader RiC-O concept; and Instance-Level Multi-Typing (`rdf:type`) for core structural objects where class subsumption is overly rigid. In instance multi-typing, individual real-world nodes in the RDF graph hold dual class assertions, satisfying both archival and preservation models simultaneously. Table 1 summarizes the finalized semantic alignment across all core entities.

4.1 Detailed Entity Relationship Analysis

The structural backbone of the alignment framework reconciles core objects through top-level subsumption and instance-level multi-typing. At the top level, `premis:Object` serves as the abstract root for items managed by a preservation system, while `rico:Thing` is the top-level abstract entity encompassing all physical, digital, or conceptual entities in RiC-O. Asserting that `premis:Object` `rdfs:subClassOf` `rico:Thing` establishes a top-level ontological anchor, ensuring that every preserved object is recognized as a valid entity within the archival universe.

For primary structural entities, subclassing is intentionally avoided to prevent schema rigidity. A `premis:IntellectualEntity` represents a distinct intellectual work, whereas `rico:RecordResource` represents content carrying archival provenance. Rather than asserting a TBox subclass relation, when an intellectual

RiC-O Entity	PREMIS v3 Entity	Type of Relationship
<code>rico:Thing</code>	<code>premis:Object</code>	<code>rdfs:subClassOf</code>
<code>rico:RecordResource</code>	<code>premis:IntellectualEntity</code>	Instance-level Multi-typing (<code>rdf:type</code>)
<code>rico:Instantiation</code>	<code>premis:Representation</code>	Instance-level Multi-typing (<code>rdf:type</code>)
<code>rico:Instantiation</code>	<code>premis:File</code>	Instance-level Multi-typing (<code>rdf:type</code>)
<code>rico:Event</code>	<code>premis:Event</code>	<code>rdfs:subClassOf</code>
<code>rico:Activity</code>	<code>premis:Event</code>	<code>rdfs:subClassOf</code>
<code>rico:Identifier</code>	<code>premis:Identifier</code>	<code>owl:equivalentClass</code>
<code>rico:ActivityType</code>	<code>premis:Action</code>	<code>rdfs:subClassOf</code>
<code>rico:CarrierType</code>	<code>premis:StorageMedium</code>	<code>rdfs:subClassOf</code>
<code>rico:Agent</code>	<code>premis:Agent</code>	<code>rdfs:subClassOf</code>
<code>rico:Person</code>	<code>premis:Person</code>	<code>rdfs:subClassOf</code>
<code>rico:CorporateBody</code>	<code>premis:Organization</code>	<code>rdfs:subClassOf</code>
<code>rico:Mechanism</code>	<code>premis:SoftwareAgent</code>	<code>rdfs:subClassOf</code>
<code>rico:Mechanism</code>	<code>premis:HardwareAgent</code>	<code>rdfs:subClassOf</code>
<code>rico:Rule</code>	<code>premis:RightsBasis</code>	<code>rdfs:subClassOf</code>
<code>rico:Rule</code>	<code>premis:PreservationPolicy</code>	<code>rdfs:subClassOf</code>
<code>rico:Rule</code>	<code>premis:Rule</code>	<code>rdfs:subClassOf</code>
<code>rico:Mandate</code>	<code>premis:Statute</code>	<code>rdfs:subClassOf</code>

Table 1. Semantic Alignment between RiC-O and PREMIS.

entity is formally ingested into a repository, the specific instance is assigned dual types (`ex:Report_2026` a `premis:IntellectualEntity`, `rico:Record`). This ensures that abstract preservation objects only receive archival status when they carry genuine provenance significance. Similarly, a `premis:Representation` represents the set of files required to render a complete work, which aligns with `rico:Instantiation` as the physical or logical realization of a record. An instance such as a digital manuscript rendering (`ex:Rep_001` a `premis:Representation`, `rico:Instantiation`) allows SPARQL queries to traverse logical structures across both descriptive and preservation domains. At the file level, a `premis:File` represents an ordered sequence of bytes, corresponding to `rico:SingleLogicalCarrier` (a direct subclass of `rico:Instantiation`). Employing instance-level multi-typing (`ex:master.tif` a `premis:File`, `rico:SingleLogicalCarrier`) prevents low-level system files, such as temporary sidecar metadata XMLs, from automatically burdening the archival descriptive hierarchy unless explicitly asserted as record instantiations.

Events, activities, and identifiers represent areas of strong semantic convergence. A `premis:Event` records an operational action performed upon a digital object, such as a checksum verification or format migration. In RiC-O, `rico:Event` represents occurrences in time and space, while `rico:Activity` represents intentional conduct by agents. Because all preservation events are

discrete occurrences that affect archival instantiations, `premis:Event` is asserted as a subclass of both `rico:Event` and `rico:Activity`. For instance, a fixity check execution (`ex:Event_101` a `premis:Event`) automatically behaves as an archival activity within provenance networks. To support controlled vocabularies, `premis:Action` (e.g., "ingest" or "normalization") is mapped via `rdfs:subClassOf` to `rico:ActivityType`. For instance, assigning `premis:Action` as an activity type enables preservation action codes to categorize archival activities seamlessly. Identifiers present an exact alignment: both standards employ a reified class model capturing values, schemes, and issuing authorities. Consequently, `premis:Identifier` and `rico:Identifier` are mapped as exact class equivalences (`owl:equivalentClass`), allowing an identifier node (`ex:ID_99`) to satisfy queries in both ontologies without translation.

The alignment of storage media and agents ensures that technical infrastructure and actors are fully integrated into archival context graphs. The class `premis:StorageMedium` defines the hardware hosting a bitstream, such as an LTO tape or solid-state array. Mapping `premis:StorageMedium` `rdfs:subClassOf` `rico:CarrierType` allows technical storage classifications to populate RiC-O carrier metadata. In terms of agency, PREMIS models actors via `premis:Agent`, with specialized subtypes `premis:Person` and `premis:Organization`. Aligning these as subclasses of their respective RiC-O counterparts (`premis:Agent` `rdfs:subClassOf` `rico:Agent`, `premis:Person` `rdfs:subClassOf` `rico:Person`, and `premis:Organization` `rdfs:subClassOf` `rico:CorporateBody`) ensures that human actors and institutions recorded in preservation logs automatically inherit archival agent status. Furthermore, PREMIS explicitly distinguishes automated non-human actors into `premis:SoftwareAgent` (such as FFmpeg or DROID) and `premis:HardwareAgent` (such as optical scanners). Subclassing both under `rico:Mechanism` preserves technical execution details while respecting RiC-O's broader agent taxonomy. For example, an automated extraction tool instance (`ex:DROID_v10` a `premis:SoftwareAgent`) automatically functions as a `rico:Mechanism` responsible for generating metadata activities.

Finally, legal permissions, intellectual property rights, and preservation policies govern repository operations. In PREMIS, these governance structures are represented through `premis:RightsBasis`, `premis:PreservationPolicy`, and `premis:Rule`. In RiC-O, `rico:Rule` serves as the broad superclass representing any principle, regulation, procedure, or policy governing conduct. Mapping all three PREMIS governance entities as subclasses of `rico:Rule` provides a unified framework for administrative and rights management. Specifically, `premis:Statute` represents legislative or regulatory obligations granting or restricting rights. Asserting `premis:Statute` `rdfs:subClassOf` `rico:Mandate` ensures that statutory preservation obligations, such as legal deposit acts (`ex:Statute_2024` a `premis:Statute`), map directly into legal mandates within archival context graphs.

5 Conclusion & Future work

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